



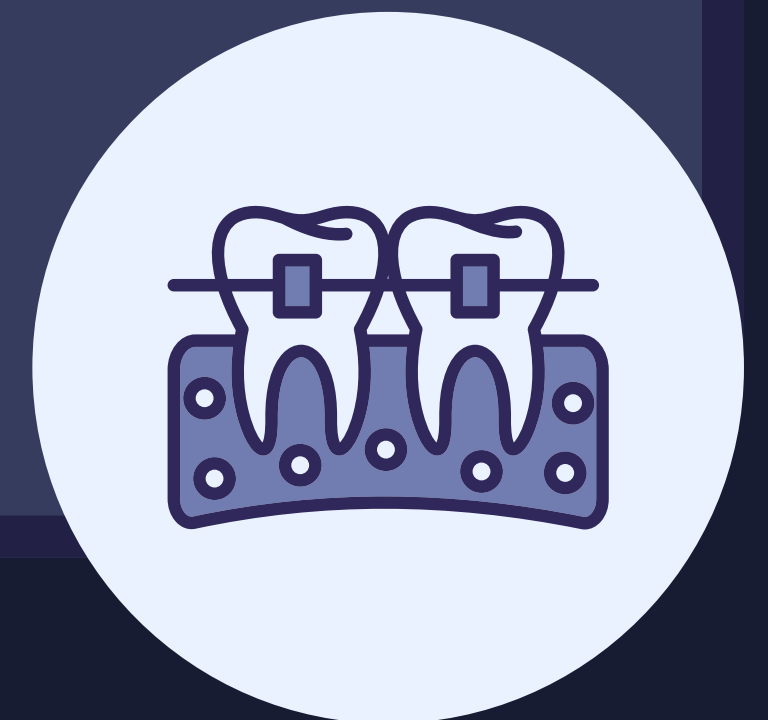
UWC Cephalometric Landmark Detection

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Yorbe Heeren



Introduction & Problem Statement

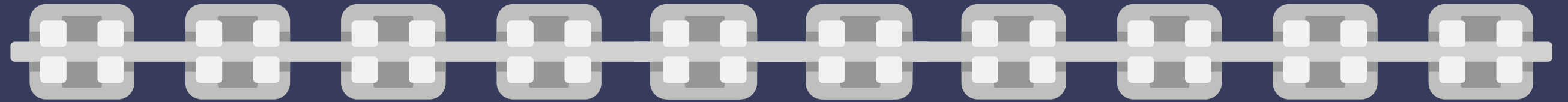
- What is Cephalometric Landmark Detection?
- Geographic Bias: European & Asian datasets exclusively
- Educational impact: unreliable training tools
- Detection tool trained specifically on South African data



Business Requirements

- Translate needs into requirements
- Clarifies problem, Defines scope and boundaries
- **8 Key Requirements:**
 - Tool Detecting 19 Anatomical Landmarks
 - Retraining / Fine-Tuning Architecture
 - Error Margin under 2 mm
 - Computational time <7 minutes
 - Ethical Compliance
 - Data Anonymisation
 - Methodological Transparency
 - Functional Prototype

Data



- Aariz
- PKU
- ISBI 2015 Grand Challenge dataset
- UWC South-African data (212 images)
- Ethics

Data Quality

- KPI=99.8% (Uniqueness)
- 100% Completeness, Consistency, Validity
- 1502 cephalometric radiographs
- 11 Duplicate

Data Processing

- File Consistency
- Image Contrasting
- **TMJ¹** Detection
- Cropping
- Heatmap Generation
- Image Resizing
- Quality Verifications

¹: Temporomandibular Joint

Figure 1: Detection of Temporomandibular Joint (TMJ)
PKU (Zeng et al., 2020)

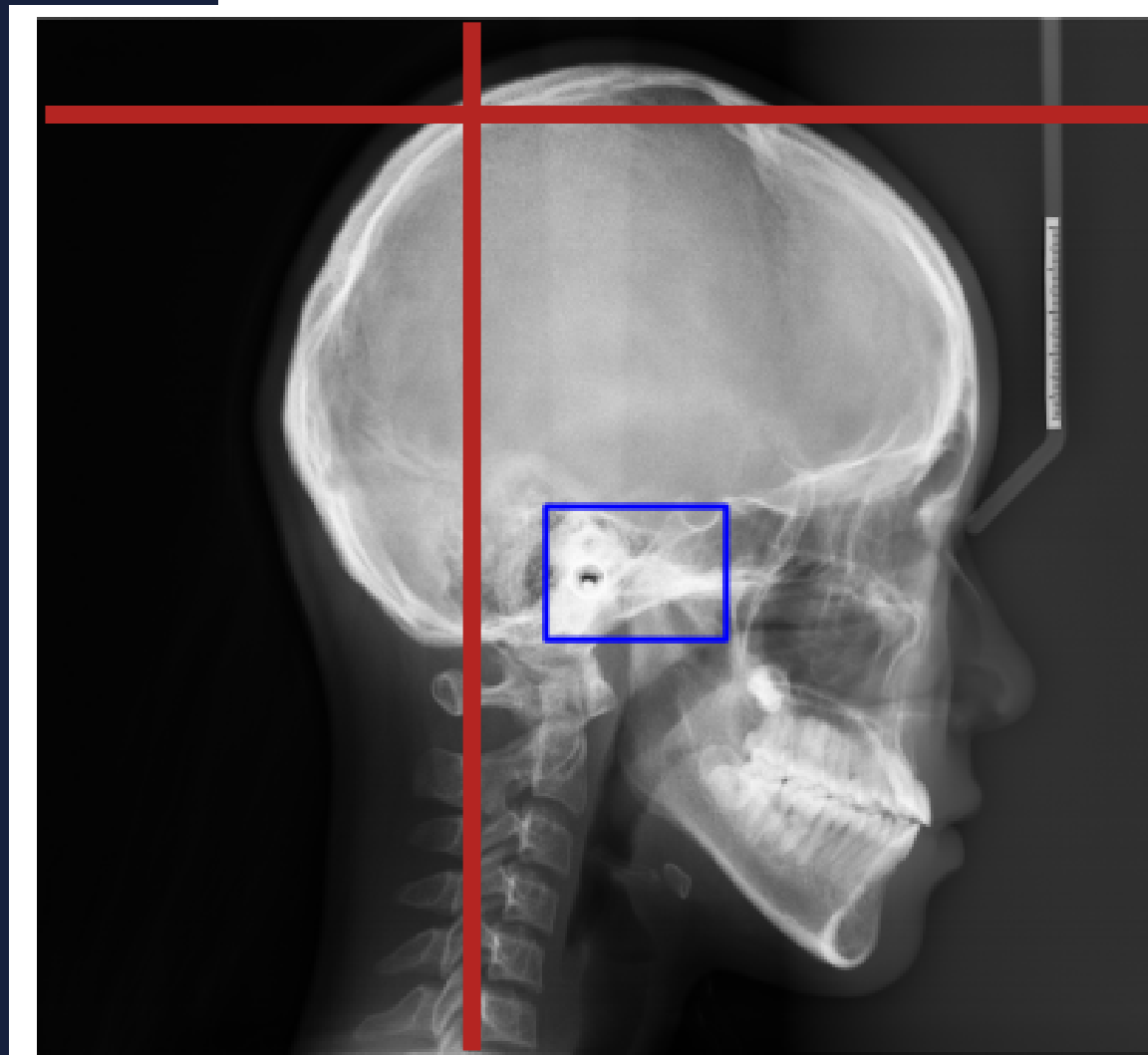


Figure 2: Cropping outcome example
PKU (Zeng et al., 2020)

Models

- Deepfuse (Aariz, PKU, Kaggle) -> 62% SDR@2mm
- Transfer Learning -> 37.8% SDR@2mm
- SA from scratch -> 81.58% SDR@2mm
- DeepFuse leveraging ALL datasets → 85.8% SDR@2mm; MRE 1.20mm

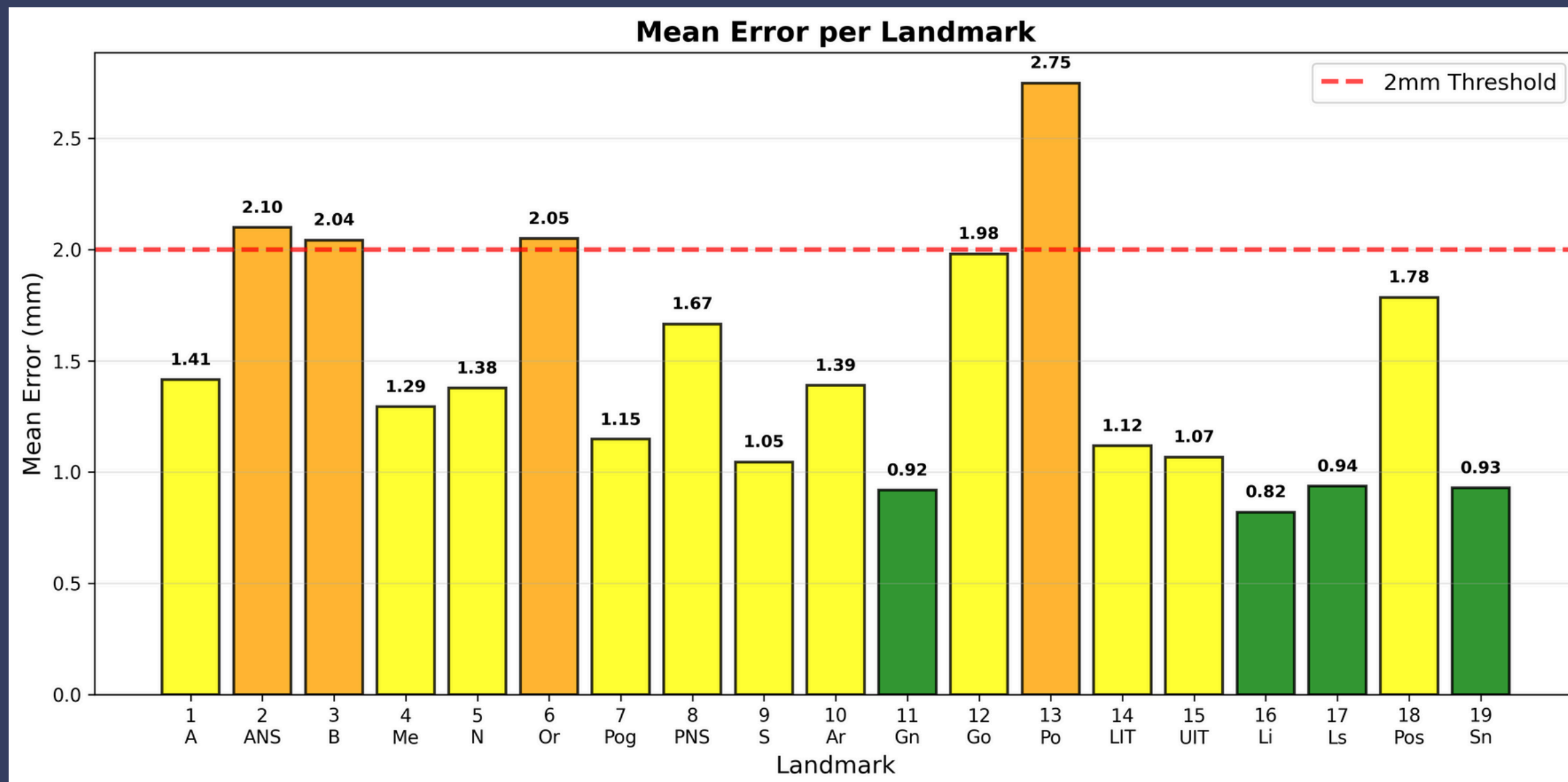
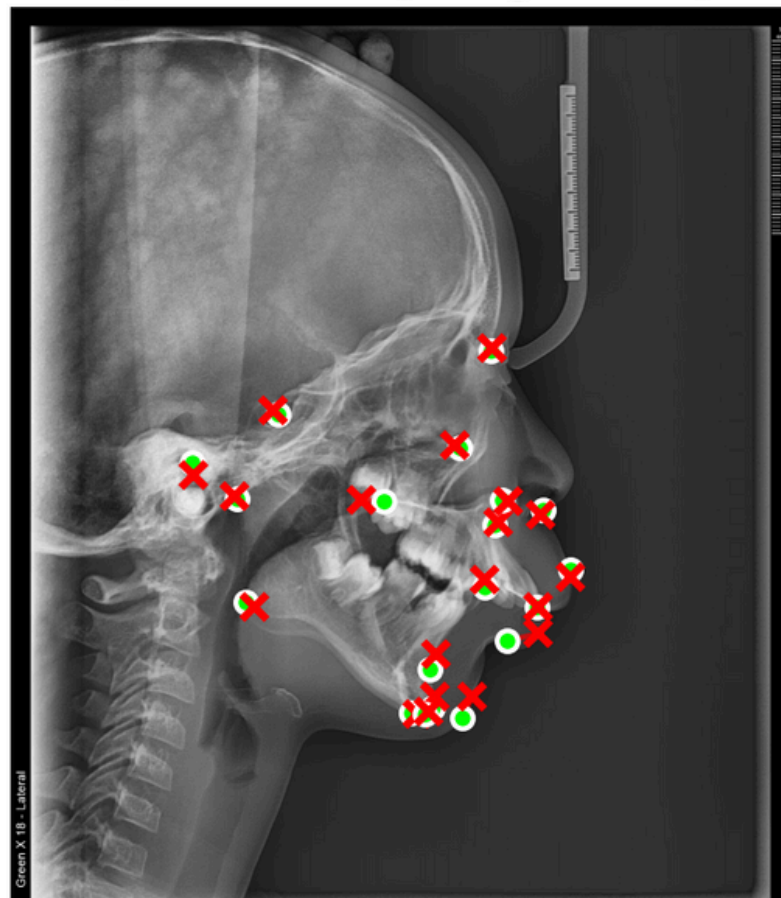
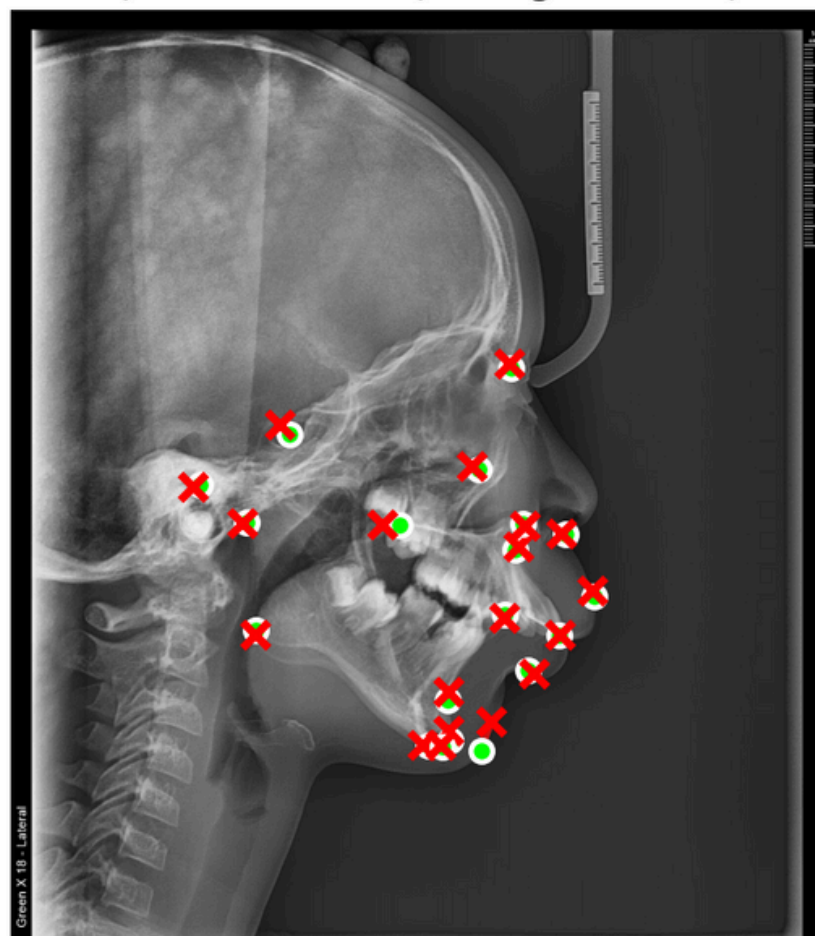


Figure 3: From scratch Model Performance on all the Landmarks

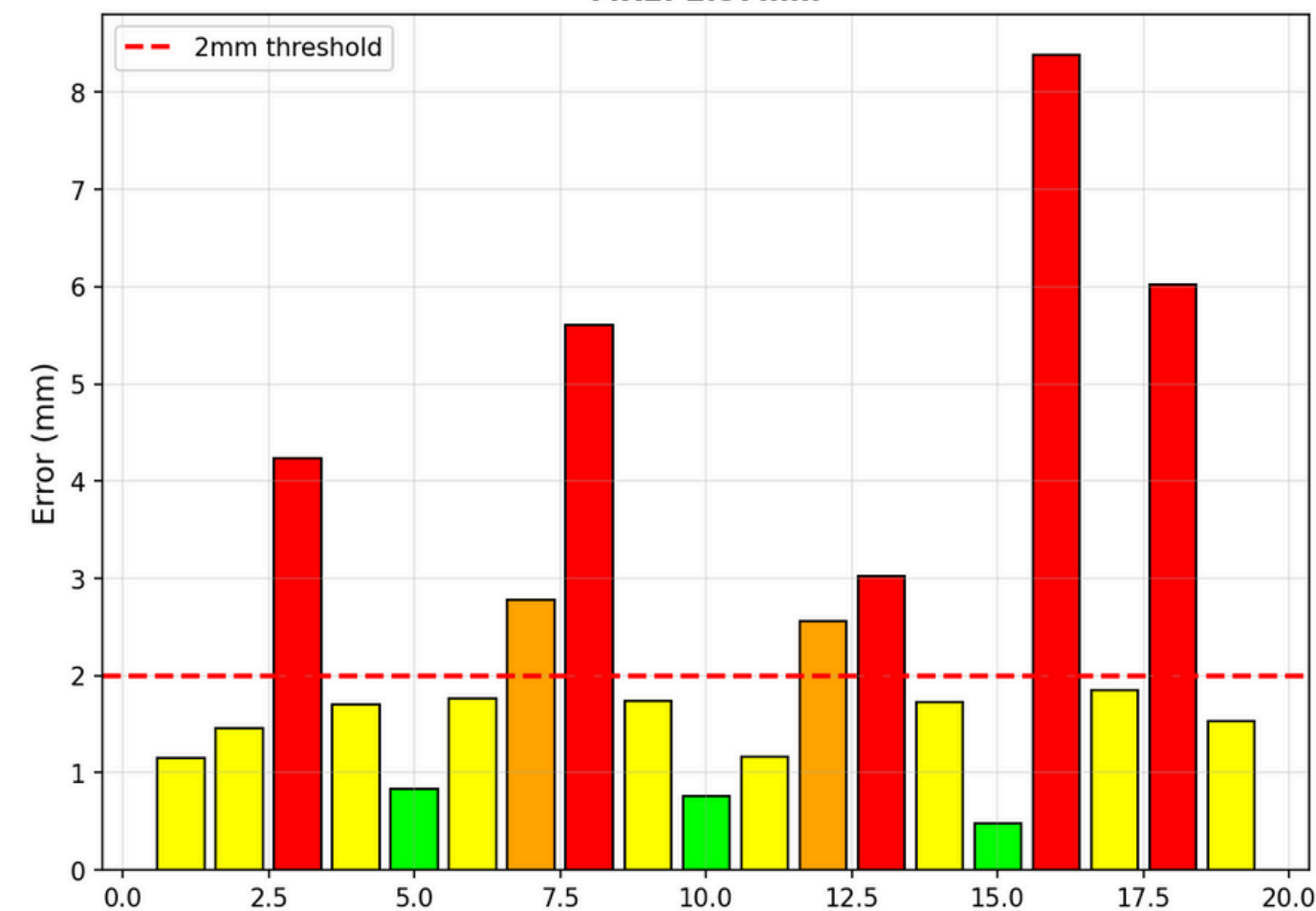
Prediction Errors
(Green=<1mm, Yellow=1-2mm, Orange=2-3mm, Red=>3mm)



Prediction Errors
(Green=<1mm, Yellow=1-2mm, Orange=2-3mm, Red=>3mm)



Per-Landmark Error
MRE: 2.57mm



Per-Landmark Error
MRE: 1.82mm

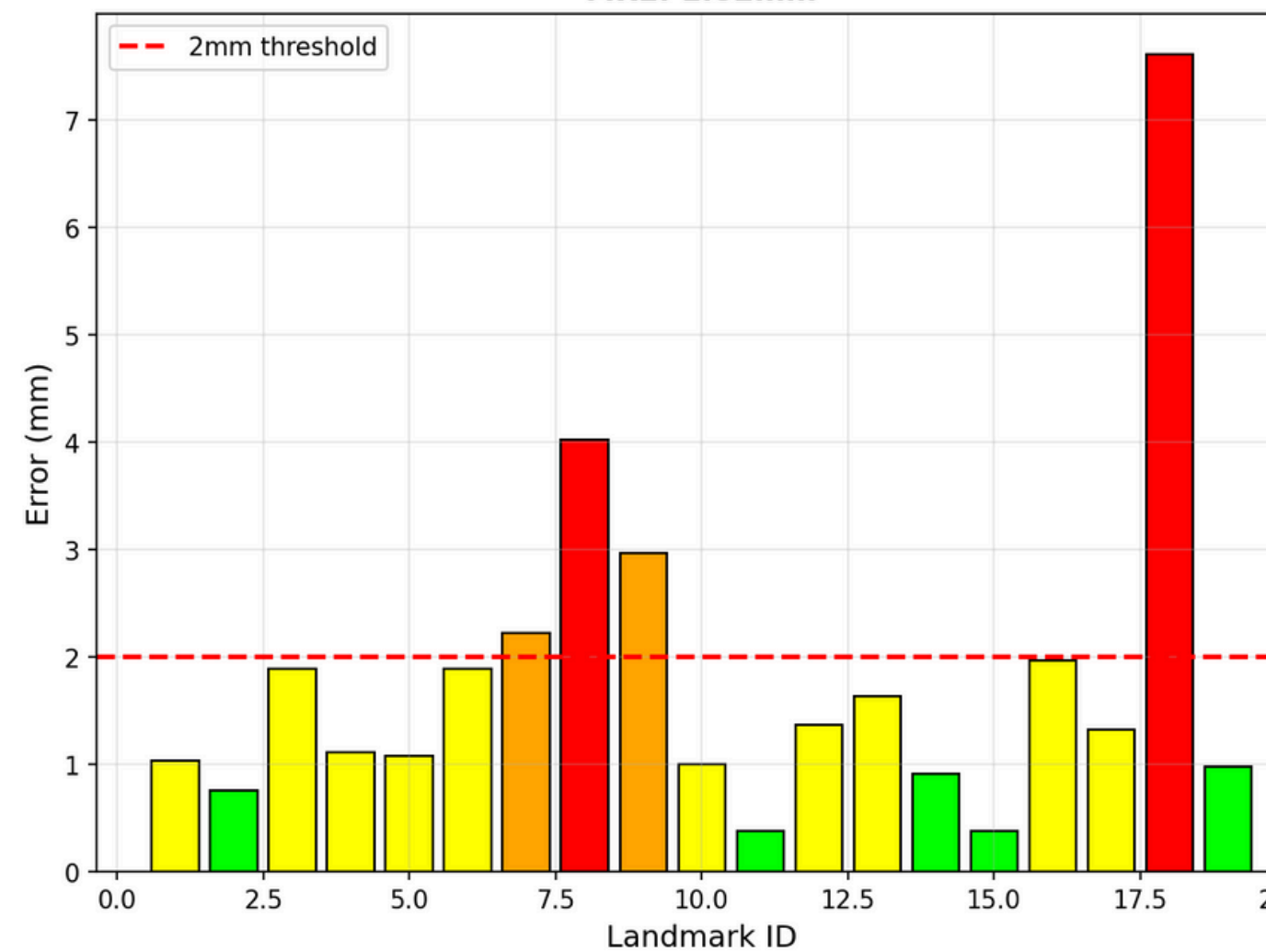


Figure 4: Baseline model Performance

The baseline model will need more human attention.

Figure 5: SA from scratch model Performance

Deployment Azure

- Azure Deployment
- Azure Container App
- Pipelines Airflow
- Blue/Green Model
- Difficult/Expensive

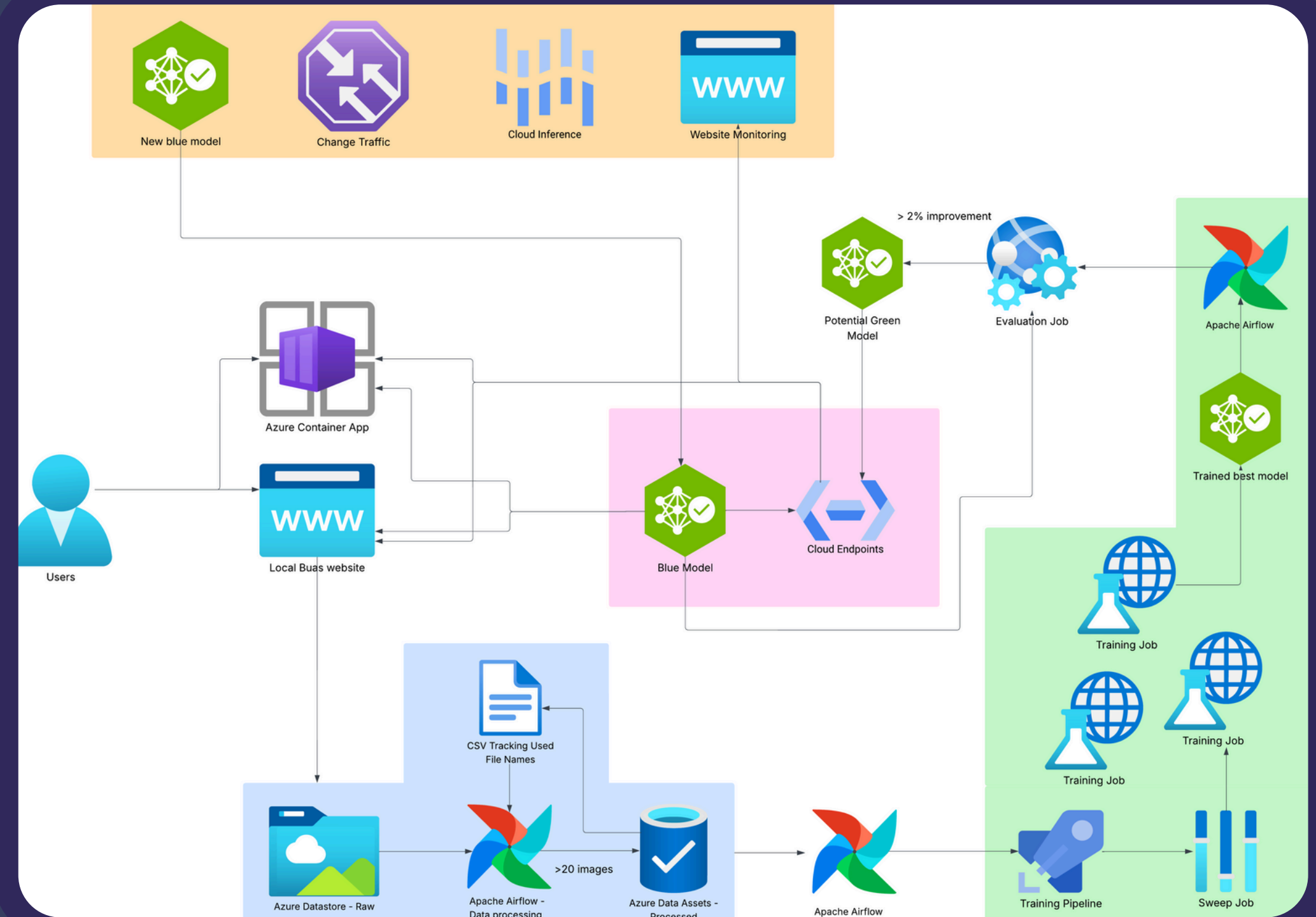


Figure 6: Azure Deployment

Deployment GCS

- GCS Deployment
- Images GCS
- Retraining Locally
- Local Deployment
- Easy/Inexpensive

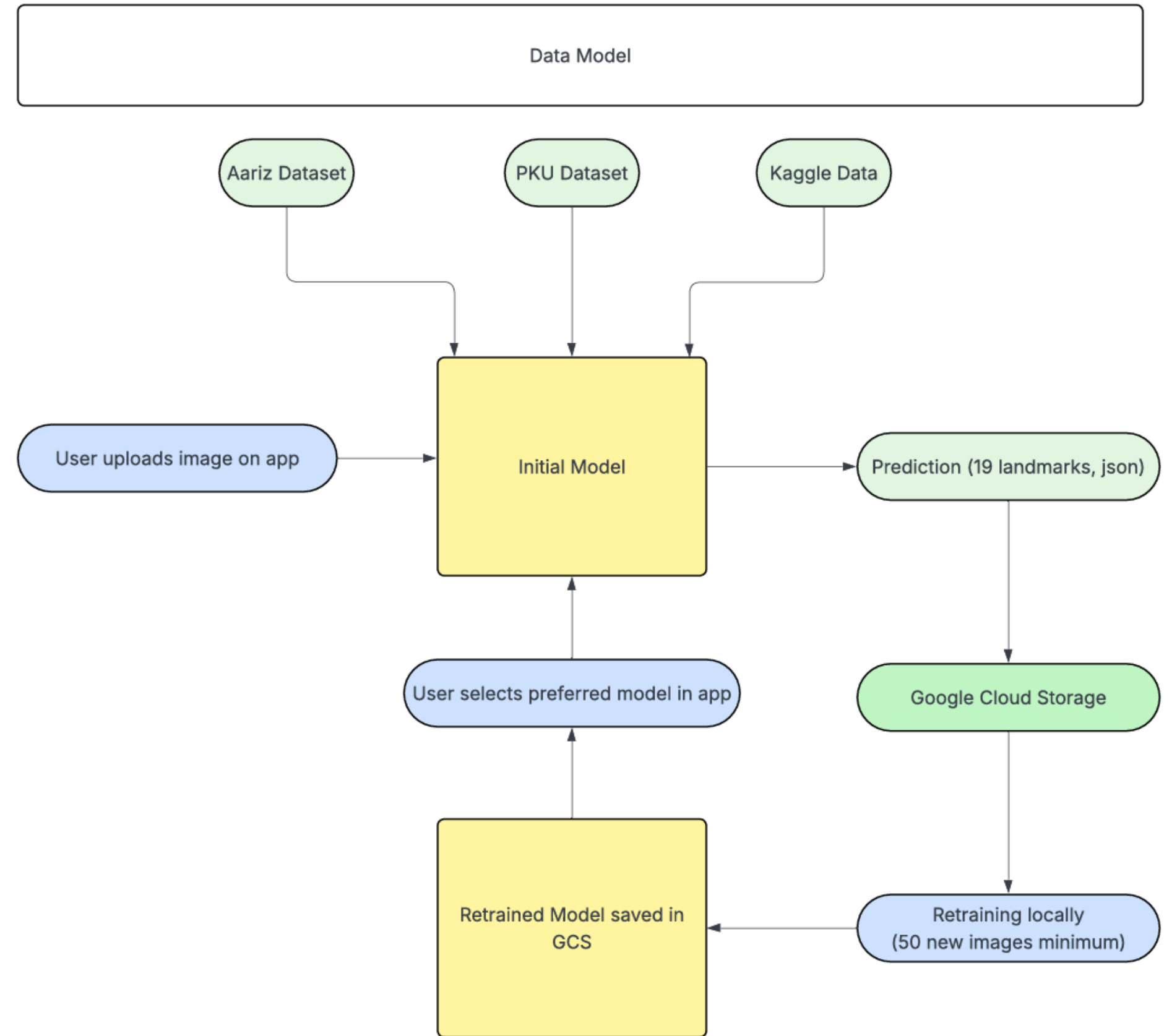


Figure 7: GCS Deployment



Demo



Limitations

- Clinical Deployment Threshold Gap
- Dataset Constraints
- Inter-Dataset Heterogeneity
- Resolution Trade-offs

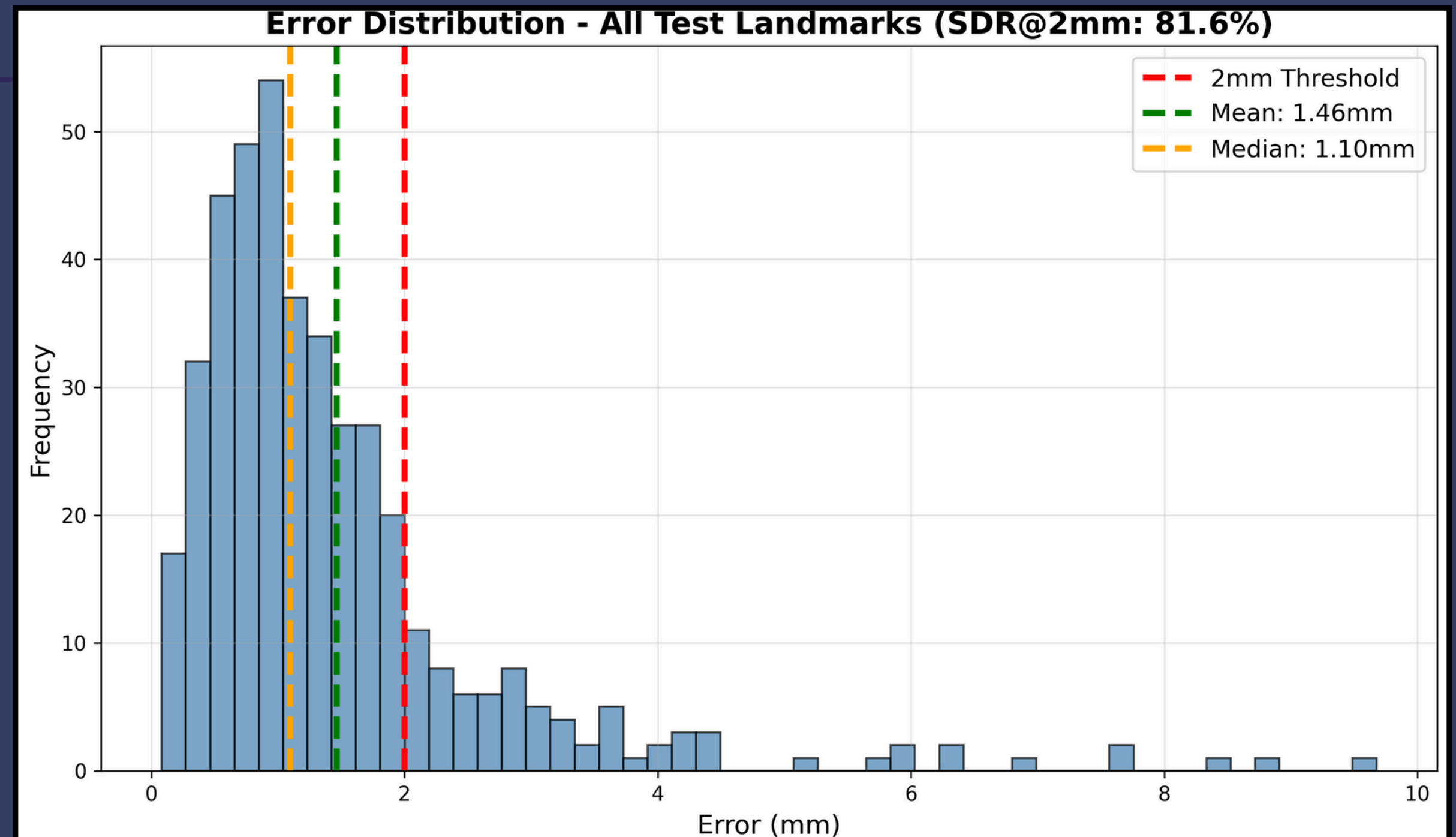


Figure 8: Error Distribution per Test Landmarks

Future Steps & Recommendations

- Expand the SA dataset.
- Recommended areas of improvement:
 - Usability;
 - Transparency;
 - Deployment.

References

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Thank You !